Propensity-Augmented Causal Foundation Models

2026-01-06

Author: Jan Teichert-Kluge, PhD Student, University of Hamburg

Recipient: Prof. Rahul G. Krishnan, University of Toronto / Vector Institute

Topic: Proposal for Visiting Research Stay (Connecting the "Hamburg School" of DML with the "Toronto School" of PFNs)

The Core Problem

Current Causal Foundation Models (like CausalPFN/CausalFM) are powerful because they learn Bayesian inference algorithms in-context. However, they typically rely on the transformer to *implicitly* reconstruct the treatment assignment mechanism ($P(A|X)$) solely from raw covariates X. This implicit approach faces theoretical risks:

1. **Inefficiency:** In high-dimensional settings, attention mechanisms struggle to identify sparse confounders without guidance.
2. **Robustness:** If the test data's selection mechanism shifts outside the training prior's support (e.g., sharper boundaries), the model has no fallback.
3. **Lack of Orthogonality:** Errors in learning the selection mechanism translate directly into causal bias, violating the principle of Neyman Orthogonality used in econometrics.

The Proposed Solution: PA-CausalPFN

We propose the **Propensity-Augmented Causal PFN**. The core concept is to explicitly inject "nuisance parameters" into the foundation model's context window.

1. **Estimation:** Before inference, we can use a robust external estimator (e.g., TabPFN or Lasso) to explicitly estimate propensity scores $\hat{\pi}_i = P(A = 1|X_i)$.
2. **Augmented Tokenization:** We feed $\hat{\pi}_i$ as a privileged token alongside X_i.
3. **Structural Attention Bias:** We modify the transformer's attention heads to include a "matching bias," penalizing attention between units with vastly different propensity scores.

$$\text{Attn}(i, j) \propto \exp\left(\frac{Q_i K_j^T}{\sqrt{d}} - \lambda \cdot |\hat{\pi}_i - \hat{\pi}_j|\right)$$

The Hypothesis: "Neural Double Robustness"

This architecture aims to induce a neuro-symbolic form of Double Robustness:

- **Scenario A (Good $\hat{\pi}$):** If the external propensity model is accurate, the attention bias forces the transformer to perform "soft matching," effectively blocking backdoor paths even if the covariate embedding is imperfect.

- **Scenario B (Bad $\hat{\pi}$):** If the propensity model is misspecified, the transformer's universal approximation capabilities allow it to ignore the noisy $\hat{\pi}$ token and revert to standard in-context learning from X (outcome modeling).

Expected Outcome: A foundation model that converges at fast parametric rates when *either* the learned representation *or* the input propensity is correct, offering the safety guarantees of econometrics with the versatility of Generative AI.

Goal of the Stay

To design, train, and validate this architecture, delivering the first **Doubly Robust Foundation Model** for tabular causal inference. This creates a synthesis between rigorous semiparametric statistics and modern deep generative modeling.